Gayatri Vidya Parishad College of Engineering Autonomous

Information Technology | Course Code: IT202

Software Engineering & Design Patterns - Laboratory Manual & Practical G

Syllabus Focus: IT202 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Software Process Models & Agile Scrum

" Syllabus Scope: Waterfall, Iterative, Spiral, V-Model, Agile manifesto, Scrum sprint planning, User sto

Unit 2: Architectural Design & SOLID Principles

" Syllabus Scope: Single Responsibility, Open-Closed, Liskov Substitution, Interface Segregation, Depende

IT202: Software Engineering & Design Patterns - Advanced Topics & Assessment

Academic Track: Information Technology - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Creational & Structural Design Patterns

" Singleton, Factory Method, Abstract Factory, Builder, Prototype, Adapter, Facade, Decorator, Composite patte

Unit 4: Behavioral Design Patterns

" Observer, Strategy, Command, State, Template Method, Iterator, Mediator patterns, Loose coupling benefits

Unit 5: Software Quality Assurance & Verification

" White-box basis path testing, Cyclomatic complexity, Black-box boundary value analysis, Equivalence partitio

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Requirement Specification and UML Use Case & Class Diagram Modeling in StarUML/PlantUML

Lab 2: Implementation of Factory and Singleton Design Patterns with Thread Safety

Lab 3: Implementation of Observer Pattern for an Event Notification Subsystem

Lab 4: Automated Unit Testing and Code Coverage Profiling using Jest/JUnit

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Erich Gamma, Richard Helm - Design Patterns: Elements of Reusable Object-Oriented Software Addison-Wesl

* Ian Sommerville - Software Engineering Pearson 10th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students